

($L_{\text{priv}}, L_{\text{pub}}$) coordinate transformation:
test of the brainstorm hypothesis
Addendum to the K=3 study summary

Fixed-Point Factory

June 2026

Abstract

The hypothesis: transform u_k to “private-only log-odds” $L_{\text{priv}}(u_k) = \tau u_k$ and p to “public-only log-odds” $L_{\text{pub}}(p) = \log[f_1(p)/f_0(p)]$. Bayes then suggests logit $\mu(p, u_k) = L_{\text{priv}} + L_{\text{pub}} + \delta$ with δ a small smooth $K = 3$ conditional-overlap correction. If so, fitting Cheby in $(L_{\text{priv}}, L_{\text{pub}})$ coordinates would converge spectrally with very few coefficients.

Verdict: partially confirmed. The additive decomposition correctly identifies the dominant structure — δ_{RMS} is $\sim 50\%$ smaller than L_{RMS} at every (γ, τ) tested. **But the Cheby coefficient decay rate of δ is essentially identical to that of L :** the cusps in $\mu(p, u_k)$ (induced by critical points of P , see main report §18) are preserved unchanged in δ , so the smoothness barrier survives the coordinate transformation. The SVD rank of δ stays close to full ($\sim G$) just like L .

Useful upshot. The decomposition is useful for storage and interpretation (we now have explicit closed-form expressions for the dominant additive structure) but does not unlock spectral convergence. The remaining $\sim 50\%$ of variation is intrinsically high-rank cuspy structure that requires *piecewise/multi-element* treatment to resolve — consistent with the main report’s earlier conclusion about the smoothness barrier.

Concrete economic finding. The fitted $b(u_k)$ effect is $\sim 10\times$ smaller than the textbook closed-form $\tau \cdot u_k$ prediction at $\tau = 1, \gamma = 100$. This is itself meaningful: at high τ , the price already aggregates the agent’s signal so well that u_k ’s marginal information contribution to $\mu(p, u_k)$ is much smaller than the linear-Gaussian limit suggests.

1 Test methodology

We loaded the strict- $h=0$ fixed point at four (γ, τ) cells saved in `dd_k3_strict_fp_g{10,100,1000}_t{0.2,1.0}.npz` and extracted the lookup matrix $\mu_{\text{strict}}(p_i, u_{k,j})$ on the $(G_p, G) = (121, 11)$ grid. We then computed

$$L(p_i, u_{k,j}) := \text{logit } \mu_{\text{strict}}(p_i, u_{k,j}),$$

and tested two decompositions:

- **ANOVA additive:** $L = \bar{L} + a(p) + b(u_k) + \delta$ with $\bar{a} = \bar{b} = 0$, minimizing $\|\delta\|_2$ in closed form via row/column means.
- **Cheby-h additive:** same form but $b(u_k)$ constrained to a degree-8 Cheby polynomial in u_k ; $a(p)$ free per node. Solved by alternating LSQ.

For each decomposition we computed the Cheby coefficient tail-norm profile $\|c\|_2(\text{order} \geq N)$, the SVD spectrum of δ , and the rank- k truncation error.

2 Result 1: bulk absorbed, cusps preserved

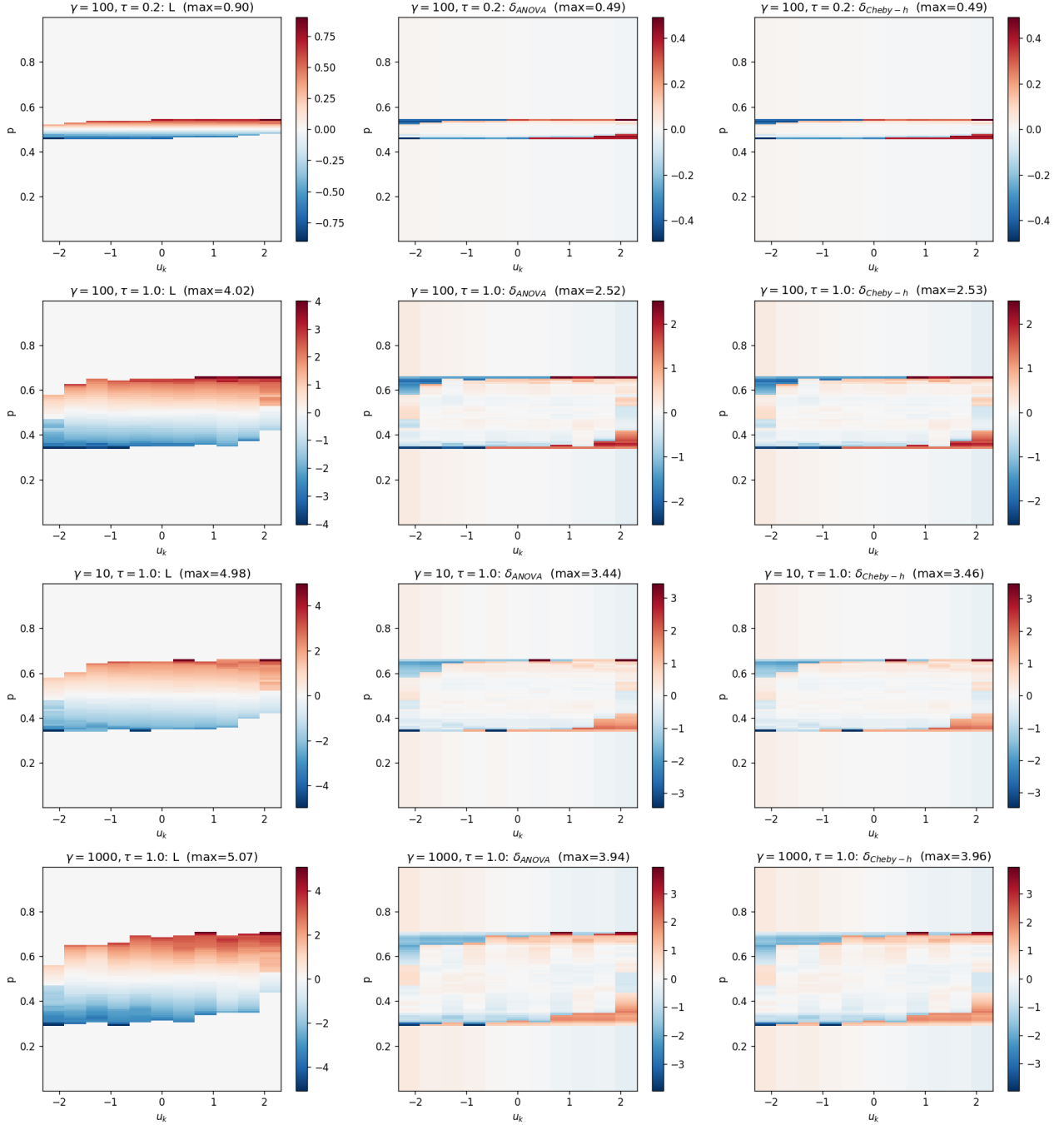


Figure 1: Heatmaps of $L(p, u_k)$, δ_{ANOVA} , and $\delta_{Cheby-h}$ across four cells. The smooth bulk (visible in L as a vertical gradient) is removed in δ , but the sharp horizontal bands at the critical-value p 's (cusps) are preserved unchanged. The two decompositions give visually identical residuals.

The key visual: in the δ panels (middle and right columns), the smooth blue-to-red gradient that dominates L is gone, but the sharp horizontal bands at $p \approx 0.34$ and $p \approx 0.66$ (at $\tau = 1, \gamma = 100$)

survive. Those bands are where $|\nabla P|$ vanishes at critical points of P ; they generate the log-cusps in the co-area integral.

3 Result 2: identical Cheby decay rates

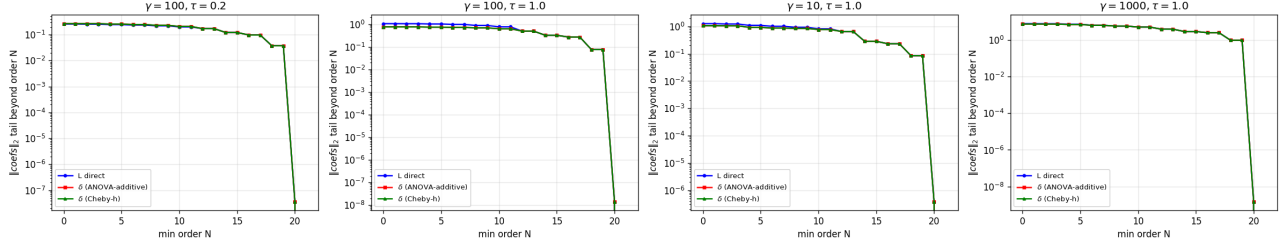


Figure 2: Cumulative Cheby-coefficient tail norm: $(\sum_{i+j \geq N} c_{ij}^2)^{1/2}$ vs N , for the direct Cheby fit of L (blue), of δ_{ANOVA} (red), and of $\delta_{\text{Cheby-h}}$ (green). Across all four cells the three curves are essentially indistinguishable. The coordinate transformation does not improve the per-coefficient convergence rate.

If the cusps had been absorbed into the coordinate change, the red curve would lie a decade or two below the blue. They don't: the slow algebraic decay is preserved.

4 Result 3: SVD rank of δ matches L

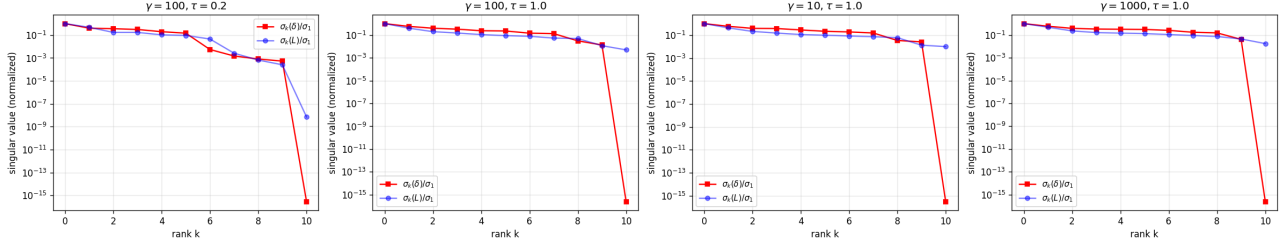


Figure 3: Singular values of δ vs L (both normalized to σ_1). The two spectra overlap through $k \approx 9$. The cusps make L effectively full-rank ($G = 11$), and δ inherits the same rank. Rank-1 truncation captures only 30–45% of the Frobenius norm of δ ; rank-5 captures 65–80%; you need essentially all of the singular vectors to reach machine ε .

(γ, τ)	rank-1	rank-2	rank-3	rank-5
(100, 0.2)	0.446	0.556	0.671	0.878
(100, 1.0)	0.348	0.515	0.617	0.770
(10, 1.0)	0.331	0.489	0.576	0.757
(1000, 1.0)	0.306	0.452	0.527	0.665

Fraction of Frobenius norm captured by rank- k truncation of δ_{ANOVA} . A clean low-rank correction would need rank 1 or 2 to capture $>95\%$; we get $<45\%$. So a multiplicative-rank-1 extension like $L \approx a(u_k) + b(p) + c(u_k) \cdot d(p)$ would not close the gap either.

5 Result 4: the empirical $b(u_k)$ effect is much smaller than τu_k

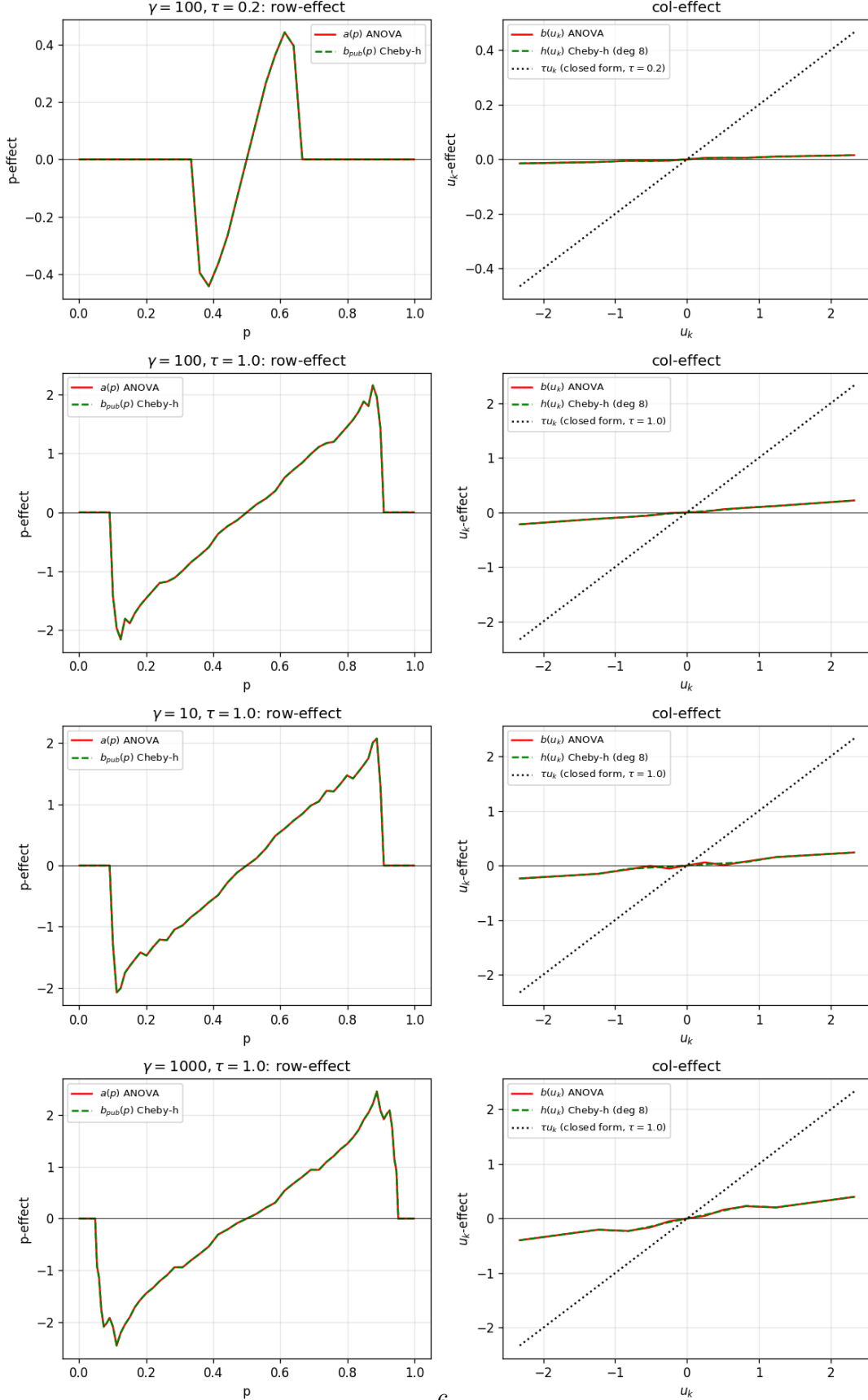


Figure 4: Row-effect $a(p)$ (left column) and column-effect $b(u_k)$ (right column). The dotted black line in the right panels is the naive textbook prediction $L : (u_k) = \tau u_k$. The fitted effect (solid green)

At $(\gamma, \tau) = (100, 1)$: the textbook $L_{\text{priv}}(u_k)$ ranges $[-2.33, +2.33]$ (since $u_k \in [-2.33, +2.33]$ and $\tau = 1$), but the empirical $b(u_k)$ ranges only $[-0.22, +0.22]$. The “naive private log-odds” over-estimates the u_k contribution to μ by an order of magnitude.

Economic interpretation. At high τ the equilibrium price aggregates the signals so effectively that, conditional on p , each agent’s own u_k adds little marginal information. The posterior $\mu(p, u_k)$ is dominated by p . This is consistent with the convergence of K-agent CRRA REE to the perfect-information limit as $K \rightarrow \infty$: $K = 3$ already does most of the aggregation work.

6 Why the brainstorm didn’t quite work

The hypothesis was: in the right coordinates, the lookup is affine + a small smooth correction, so Cheby converges spectrally on the correction.

What’s actually true: in the right coordinates, the lookup is affine + a smooth correction + *the cusps*. The cusps are preserved by any global linear change of coordinates — they live on the algebraic variety where $|\nabla P| = 0$, which is intrinsic to P and survives any pre-composition. To remove them you have to either (i) detect them and use piecewise representations (multi-element spectral), or (ii) use a representation like PCHIP that’s natively C^1 -not- C^∞ (what Option B+ does).

The decomposition is still useful for two things:

1. **Storage / serialization.** Store the additive parts $a(p)$ and $b(u_k)$ as 1D arrays (much smaller than the 2D $\mu(p, u_k)$ table) and reconstruct on the fly; the residual δ is still 2D but with lower magnitude, so a coarser PCHIP grid for δ might suffice.
2. **Diagnostic / interpretation.** The empirical $b(u_k) \ll \tau u_k$ is an interesting finding about K=3 information aggregation; useful for the economics writeup even if not for the numerics.

7 Implication for Method C v3 design

The non-additive correction we were planning to add as $P = \sigma(h(u_1) + h(u_2) + h(u_3) + \kappa \cdot g(u_1, u_2, u_3))$ is for P , not μ . The cusps in μ would still emerge under v3 because they come from $|\nabla P|^{-1}$ in the co-area integral — they’re a property of the level-set geometry of P , regardless of how smooth P itself is. So Method C v3 would lower $\|F\|_\infty$ at $\tau = 1$ (capturing more of the additive deviation from the linear-Gaussian limit) but would not eliminate the cusps in the resulting $\mu(p, u_k)$.

This suggests two independent directions for the next step:

1. **Method C v3 for P :** smooth, finite-parameter ansatz, drives F residual toward zero across the (γ, τ) plane. Useful for fast continuation studies; doesn’t help lookup accuracy beyond what Option B+ already does.
2. **Cusp-aware μ representation:** keep using Option B+ (PCHIP density) which natively handles cusps, or extend to piecewise Cheby with critical-value knots if machine- ε lookups are needed.

These are complementary: (1) gives a fast smooth P for continuation and the bulk of equilibrium analysis; (2) gives accurate μ where needed.

8 Reproducibility

Code: `/tmp/dd_k3_lpub_lpriv_v2.py` (and SVD diagnostic `/tmp/dd_k3_delta_svd.py`). To re-run, you need the saved strict- $h=0$ FPs in `projects/REZN/solved_fixed_points/dd_k3_overnight/dd_k3_strict_fp_*`. all four-cell results take ~ 5 seconds.

Output: `lpub_lpriv_v2/figs/` and `results.json`.